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If we take $x_1 = 2$ and $x_2 = 2.25$, then $y_1 = 4$ and $y_2 = 5.0625$

Here $\Delta x = 0.25 = \frac{1}{4}$, $\Delta y = 1.0625$

$$\frac{\Delta y}{\Delta x} = \frac{5.0625 - 4}{0.25} = \frac{(5.0625 - 4)}{\frac{1}{4}}$$
$$= 4(5.0625 - 4) = 4.25$$

If we take $x_1 = 2$ and $x_2 = 2.1$, then $y_1 = 4$ and $y_2 = 4.41$

Here $\Delta x = 0.1 = \frac{1}{10}$ and

$$\frac{\Delta y}{\Delta x} = \frac{(4.41 - 4)}{\frac{1}{10}} = 10(4.41 - 4) = 4.1$$

These results are tabulated as shown below:

| $x_1$ | $x_2$ | $\Delta x$ | $y_1$ | $y_2$ | $\frac{\Delta y}{\Delta x}$ |
|---|---|---|---|---|---|
| 2 | 2.25 | 0.25 | 4 | 5.0625 | 4.25 |
| 2 | 2.1 | 0.1 | 4 | 4.41 | 4.1 |
| 2 | 2.01 | 0.01 | 4 | 4.0401 | 4.01 |
| 2 | 2.001 | 0.001 | 4 | 4.004001 | 4.001 |
| 2 | 2.0001 | 0.0001 | 4 | 4.00040001 | 4.0001 |

From the above table, the following inferences can be made.

- As $\Delta x$ tends to zero, $\frac{\Delta y}{\Delta x}$ approaches the limit given by the number 4.
- At a point x = 2, the derivative $\frac{dy}{dx} = 4$.
- It should also be mentioned here that $\Delta x \to 0$ does not mean that $\Delta x = 0$.

This is because, if we substitute $\Delta x = 0$, $\frac{\Delta y}{\Delta x}$ becomes indeterminate.

In general, we can obtain the derivative of the function $y = x^2$, as follows:

$$\frac{\Delta y}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x} = \frac{x^2 + 2x\Delta x + \Delta x^2 - x^2}{\Delta x}$$
$$= \frac{2x\Delta x + \Delta x^2}{\Delta x} = 2x + \Delta x$$
$$\frac{dy}{dx} = \lim_{\Delta x \to 0} 2x + \Delta x = 2x$$

The table below shows the derivatives of some common functions used in physics

| Function | Derivative |
|---|---|
| $y = x$ | $dy/dx = 1$ |
| $y = x^2$ | $dy/dx = 2x$ |
| $y = x^3$ | $dy/dx = 3x^2$ |
| $y = x^n$ | $dy/dx = nx^{n-1}$ |
| $y = \sin x$ | $dy/dx = \cos x$ |
| $y = \cos x$ | $dy/dx = -\sin x$ |
| $y =$ constant | $dy/dx = 0$ |
| $y = AB$ | $\frac{dy}{dx} = A\left(\frac{dB}{dx}\right) + \left(\frac{dA}{dx}\right)B$ |

In physics, velocity, speed and acceleration are all derivatives with respect to time 't'. This will be dealt with in the next section.

## EXAMPLE 2.19

Find the derivative with respect to t, of the function $x = A_0 + A_1 t + A_2 t^2$ where $A_0$, $A_1$ and $A_2$ are constants.

### Solution

Note that here the independent variable is 't' and the dependent variable is 'x'